

GUIDE SPECIFICATION FOR VETPROOF UM765 LIQUID-APPLIED WATERPROOFING MEMBRANE

SECTION 07 14 16

COLD-FLUID APPLIED WATERPROOFING

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Surface preparation.
- B. Application of single-component, cold-applied, liquid waterproofing membrane.

1.02 RELATED SECTIONS

- A. Section 03 30 00 – Cast-in-Place Concrete.
- B. Section 04 20 00 – Unit Masonry.
- C. Section 07 21 00 – Thermal Insulation.
- D. Section 07 60 00 – Flashing and Sheet Metal.
- E. Section 07 92 00 – Joint Sealants.
- F. Section 33 46 13 - Foundation Drainage.

1.03 REFERENCES

- A. ASTM C836 - Standard Specification for High Solids Content, Cold Liquid-Applied Elastomeric Waterproofing Membrane for Use with Separate Wearing Course.
- B. ASTM C661 - Standard Test Method for Indentation Hardness of Elastomeric-Type Sealants by Means of a Durometer.
- C. ASTM D816 - Standard Test Methods for Rubber Cements.
- D. ASTM D1644 - Standard Test Methods for Nonvolatile Content of Varnishes.
- E. ASTM D2370 - Standard Test Method for Tensile Properties of Organic Coatings.
- F. ASTM D2697 - Standard Test Method for Volume Nonvolatile Matter in Clear or Pigmented Coatings.
- G. ASTM E96 - Standard Test Methods for Water Vapor Transmission of Materials.

1.04 SUBMITTALS

- A. Comply with Section 01 33 00 - Submittal Procedures.
- B. Submit manufacturer's product data and application instructions.

1.05 QUALITY ASSURANCE

- A. Installer Qualifications:
 - a. Use an experienced installer and adequate number of skilled personnel who are thoroughly trained and experienced in the application of fluid applied waterproofing membranes.
- B. Obtain waterproofing materials from a single manufacturer regularly engaged in manufacturing the product.
- C. Provide products which comply with all state and local regulations controlling use of volatile organic compounds (VOCs).

1.06 PRECONSTRUCTION MEETING

- A. Preconstruction Meeting: Convene one week prior to commencing work of this section, in accordance with Section () - Project Meetings.

1.07 MOCK-UPS

- A. Prior to installation of waterproofing membrane, apply waterproofing membrane to 100 ft.² of deck or wall to demonstrate surface preparation, crack and joint treatment, corner treatment, thickness, and to demonstrate tie-ins with adjoining construction, and other termination conditions, as well as qualities of materials and execution.
- B. Cooperate and coordinate with the owner's inspection and testing agency. Do not cover any installed waterproofing membrane unless it has been inspected, tested and approved.

1.08 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials to site in manufacturer's original, unopened containers and packaging, with labels clearly identifying product name and manufacturer.
- B. Store materials in a clean, dry area in accordance with manufacturer's instructions.
- C. Store at temperatures below 35 °C.
- D. Protect materials during handling and application to prevent damage or contamination.

1.09 ENVIRONMENTAL REQUIREMENTS

- A. Product not intended for uses subject to abuse or permanent exposure to the elements.
- B. Do not apply membrane when air, material, or surface temperatures are expected to fall below 4° C) within four hours of completed application.
- C. Do not apply membrane if rainfall is forecast or imminent within 12 hours.
- D. Do not apply waterproofing membrane to any surfaces containing frost.
- E. Consult manufacturer for applications to green concrete.

PART 2 PRODUCTS

2.01 MANUFACTURER

- A. Saudi Vetonit Co. Ltd.-Saveto. P.O.Box 52235 - Riyadh 11563 Saudi Arabia. Tel: +966 11 265 3334 .Fax: +966 11 265 0936 ,www.saveto.com

2.02 MATERIALS

- A. Waterproofing Membrane: single-component, cold-applied. Premium quality pitch extended polyurethane waterproofing membrane.
 - 1. Performance Based Spec: Waterproofing membrane shall have the following properties as determined by laboratory testing:
 - a. Elongation ASTM D412-06a (2013) :> 600 %
 - b. Tensile Strength: ASTM D412-06a (2013): 2MPa
 - c. Tear Resistance ASTM D624-00 (2012): 10 KN/m
 - d. Water Vapour Transmission ASTM E96/E96M-13: 9.6 (gm / m² / 24 hrs.)
 - e. Shore A Hardness ASTM D2240-05 (2010): > 32
 - f. Elastic Recovery %: ASTM D412-06a (2013): 95
 - 2. Proprietary Based Spec:
 - a. Vetoproof UM765 Waterproofing Membrane Vertical grade & Horizontal grade manufactured by Saudi Vetonit Co. Ltd-Saveto

2.03 ACCESSORIES

- A. Saveto Concrete Repair Materials
- B. Polypropylene Protection Board
- C. Polyethylene Protection Board

PART 3 EXECUTION

3.01 EXAMINATION

- A. Examine surfaces to receive membrane. Notify architect if surfaces are not acceptable. Do not begin surface preparation or application until unacceptable conditions have been corrected.

3.02 SURFACE PREPARATION

- A. Protect adjacent surfaces not designated to receive waterproofing.
- B. Clean and prepare surfaces to receive waterproofing in accordance with manufacturer's instructions.
- C. Do not apply waterproofing to surfaces unacceptable to manufacturer.
- D. Clean concrete surfaces so they are free of all coatings, dirt, oil, paints and any other contaminants.
- E. Patch all holes and voids and smooth out any surface misalignments.
- F. Remove and patch all concrete form ties.
- G. Treatment of Existing Cracks and All Non-Structural Joints
 - 1. Identify and install detailing membrane in all cracks and all non-structural joints.
 - 2. Apply a 30 wet mil coat of the fluid applied membrane ensuring that there is a minimum of 75 mm of membrane extending onto the wall in all directions.
 - 3. Embed the non-woven reinforcing fabric over the entire area of this membrane and work in using trowel.
 - 4. Completely cover the glass mesh with a second coat of the fluid applied membrane at 30 wet mils while the first coat is still wet, again extending 75mm onto the wall in all directions.

H. Treatment of Inside & Outside Corners

1. Install detailing membrane to create a minimum $\frac{3}{4}$ " fillet in all inside corners.
2. Apply a 30 wet mil coat of the fluid applied membrane ensuring that there is a minimum of 75 mm of membrane extending onto the wall in all directions.
3. Embed the non-woven reinforcing fabric over the entire area of this membrane and work in using trowel.
4. Completely cover the glass mesh with a second coat of fluid applied membrane at 30 wet mils while the first coat is still wet, again extending 75mm onto the wall in all directions.
5. On outside corners subject to backfilling, install reinforced joint tape in lieu of fabric joint tape following the same procedure.

3.03 APPLICATION

- A. Apply waterproofing membrane system in accordance with manufacturer's instructions.
- B. Gently mix membrane prior to application.
- C. Apply membrane by brush, trowel or a gaged spreader at a minimum thickness of 1 mm.
- D. If a two-coat application is required, apply second coat as soon as possible with no more than eight hours between coats providing a minimum total thickness of 60 wet mils.
- E. Frequently inspect surface area to ensure proper adhesion and consistent thickness is achieved.
- F. Work material into any fluted rib forming indentations.
- G. Provide minimum cured membrane thickness of 60 mils dry.

3.04 PROTECTION

- A. Protect membrane with application of waterproofing protection course, drainage board, or other approved material.
- B. Backfill immediately using care to avoid damaging waterproofing membrane system.

END OF SECTION